

Coding & Solution Specification

Date	06 July 2026	Project Name	Smart Lender
Team Lead	Alla Umesh	Team Members	DSP, JRJ, LPB, SVB

Solution Summary Details	
Programming Language	Python 3.13+
Frameworks Used	Flask web server framework
Machine Learning Model	XGBoost Classifier (Best Model serialized in rdf.pkl)
Key Features Implemented	1. Instantly predict single applicants eligibility. 2. Evaluate batch applicant spreadsheets asynchronously. 3. Automatic risk evaluation and flagging. 4. Light-mode metrics landing dashboard.
Pending Features	User database authentication (future roadmap)
Repository Link / URL	https://github.com/Umeshh27/smart-lender

Code Quality Checklist			
S.No	Criteria	Status (Yes/No)	Comments
1	Code modularized into functions	Yes	Clear logical functions in train.py/app.py
2	Descriptive naming used	Yes	Follows standard clean coding patterns
3	Comments explaining operations	Yes	Comments trace preprocessing mapping
4	Error handling implemented	Yes	Graceful feedback alerts on failures
5	Application runs without crashes	Yes	Tested local boot of Flask web app
6	Committed to Git version control	Yes	Tracked under github repo Umeshh27/smart-lender